

PAPER Recirculation-Resolved Q_E Closure in a D– ^3He Tandem-Mirror Burner: The Synchrotron-Limited Operating Point and the Carried-Open Plug Gate

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Abstract

A power-producing magnetic mirror is judged not by its physics gain but by its *recirculation-resolved* engineering gain $Q_E \equiv P_{\text{gross}}/P_{\text{recirc}}$, the ratio of gross electrical output to the electrical power that must be pumped back into the plug and thermal-barrier drive to keep it burning. We report a zero-dimensional, self-consistent power balance for the Kronos D– ^3He tandem-mirror burner — a Pastukhov-plugged central cell shared by the Aegis fixed-installation and MetroVolt data-centre products — in which every term of the recirculating loop is resolved. At the frozen design point the balance closes at $Q_E = 1.318$, with fusion power $P_{\text{fus}} = 4298.5$ MW, neutron power $P_n = 233.95$ MW and a neutron fraction of only $f_n = 5.44\%$, at a self-consistent central electron temperature $T_e = 89.72$ keV and ambipolar ion-confining potential $\varphi_i = 248.75$ keV. Resolving the plant loop for the full-scale reference gives a gross electrical output of 3522 MWe against a recirculated drive of 2672 MWe — a recirculating fraction of $0.759 = 1/Q_E$ — for a net export of +850 MWe. We derive the governing model from the fusion, radiation and Pastukhov end-loss balances, cast the engineering gain and its net-power constraint as explicit equations, and show that the operating point is set, above all else, by the synchrotron radiative-transfer model: the correct reflective-cavity (Albajar–Fidone–Johner, AFJ) treatment gives $Q_E = 1.32$, an optically-thin wall collapses the gain to $Q_E = 0.39$, and net gain therefore imposes a hard first-wall specification, $R_{\text{wall}} \gtrsim 0.83$. The closure is robust: a 40,000-sample global uncertainty quantification on the NVIDIA GPU HPC campaign returns $Q_E = 1.317 \pm 0.14$ with $P(Q_E > 1) = 1.0$, a 25-case design-space scan clears unity in every configuration, and a net-power Monte-Carlo places the machine at $P(\text{net} > 0) = 73\%$ when it is deliberately pushed to the cliff. A Saltelli–Sobol decomposition names the single load-bearing lever: the plug-to-central density ratio ($S_1 = 0.41$, $S_T = 0.87$), whose net-gain floor sits at $n_p/n_c = 14$ against a design value of 16. This is the *carried-open plug gate* — the one honest open engineering item — and it is reported here as an explicit, testable requirement rather than a hidden assumption. The reported gain uses a deliberately conservative direct-energy-conversion efficiency, $\eta_{\text{DEC}} = 0.49$, below the literature floor of the DEC ladder, so the closure is a floor and not a ceiling. The burner’s binding field is the 26.49 T plug coil, a magnet distinct from the Hyperion breeder centrepost.

Keywords: tandem mirror; D– ^3He fusion; synchrotron radiation; recirculating power; engineering gain; Pastukhov plug; direct energy conversion; uncertainty quantification

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1 Introduction

1.1 The mirror power balance and the aneutronic promise

A power-producing magnetic mirror lives or dies by its power balance. Unlike a toroidal device, a linear tandem mirror is an inherently *open* system: particles and energy leak continuously out of the ends, and the machine is held in a burning state only by a large, continuously supplied plug and thermal-barrier drive. The figure of merit for such a plant is therefore not the physics gain $Q = P_{\text{fus}}/P_{\text{heat}}$ but the *recirculation-resolved engineering gain*,

$$Q_E \equiv \frac{P_{\text{gross}}}{P_{\text{recirc}}}, \quad (1)$$

the ratio of gross electrical output to the electrical power recirculated into the drive; net electrical export is $P_{\text{net}} = P_{\text{gross}} - P_{\text{recirc}} = P_{\text{recirc}}(Q_E - 1)$, and the machine is a power plant only if $Q_E > 1$. A mirror that closes its physics balance can still fail as an engineering system if the recirculating loop is too large; conversely, resolving that loop term-by-term is what turns a favourable single-point balance into a buildable specification.

The Kronos burner is a D–³He tandem mirror. Its promise is a largely aneutronic reaction: the dominant $\text{D} + {}^3\text{He} \rightarrow {}^4\text{He} + p$ channel releases 18.35 MeV entirely into charged products, so most of the fusion power emerges as fast ions and protons that a direct energy converter (DEC) turns into electricity without a thermal cycle [1, 2, 3]. The neutron budget is set only by parasitic D–D and by the D–T self-burn of the tritium those D–D reactions make; the neutron fraction is consequently a small number, which is what makes the burner sitable near defended installations and data centres [4, 5]. The counterpart risk is that the very high electron temperature that lifts the D–³He reactivity also drives strong radiation: bremsstrahlung and, above all, synchrotron emission from a hot magnetized electron population [6, 8]. The burner’s operating point is thus a competition between an aneutronic charged-power source and a radiation sink, mediated by an end-loss channel that a strong electrostatic plug must suppress.

1.2 Prior art

The tandem mirror is a mature reactor concept. The end-plug idea of Dimov and of Fowler and Logan [9, 10], the thermal-barrier refinement of Baldwin and Logan [11], and the Pastukhov theory of collisional end-loss from an electrostatically confined species [12, 13] together define the physics we reduce to a zero-dimensional balance here. Later work established axisymmetric MHD-stable mirror configurations [14] and simpler high-gain tandem schemes [16, 15]. The contemporary experimental basis is strong and directly relevant to the Kronos field regime: the gas-dynamic trap raised bulk electron temperatures into the several-hundred-eV range [22, 23], and the Wisconsin HTS Axisymmetric Mirror (WHAM) now operates a 17 T high-temperature-superconductor mirror — the exact magnet class the Kronos design assumes [24, 25]. On the radiation side, the reflective-cavity treatment of electron-cyclotron (synchrotron) radiative transfer by Albajar, Bornatici and Engelmann [18, 19, 20] and the universal-coefficient treatment of Trubnikov [21] bracket the emission model; the D–³He reactivities are taken from the Bosch–Hale parametrization [17]; and direct conversion follows the Venetian-blind and travelling-wave lineage of Post, Moir and Barr [1, 2, 3].

Within the Kronos series this paper is the load-bearing power-balance node. The Aegis and MetroVolt product papers state the closure as a product requirement [DOI:10.5281/zenodo.22645693; <https://www.kronosfusionenergy.com/publications/aegis-burner.html>]; the direct-energy-conversion basis is developed separately [DOI:10.5281/zenodo.22645700; <https://www.kronosfusionenergy.com/publications/direct-energy-conversion.html>]; the design-space map and the WHAM benchmark are given in the tandem-mirror mapping paper [DOI:10.5281/zenodo.22645911; <https://www.kronosfusionenergy.com/publications/tandem-mirror-design-space-mapping-for-a-d-he-burner-0-.html>]; the synchrotron transport that pins the electron temperature is treated in detail [DOI:10.5281/zenodo.22645772; <https://www.kronosfusionenergy.com/publications/synchrotron-radiation-transport.html>]; and the ECH thermal barrier and the sloshing-ion ambipolar potential that sustain the plug are the subjects of two further papers [DOI:10.5281/zenodo.22645781; <https://www.kronosfusionenergy.com/publications/ech-thermal-barrier-decoupling.html>], [DOI:10.5281/zenodo.22645913; <https://www.kronosfusionenergy.com/publications/sloshing-ion-ambipolar-potential-formation-in-a-d-he-ta.html>].

1.3 Contribution

We make the recirculating loop explicit. Prior Kronos treatments reported the physics anchors of the balance; here we (i) derive the zero-dimensional model as a coupled fixed-point system in the central-cell state, the plug potential and the electron temperature; (ii) define the recirculation-resolved engineering gain (1) and resolve every branch of the plant electrical ledger against reproduced numbers; (iii) show quantitatively that the operating point is *synchrotron-limited*, so that the first-wall reflectivity $R_{\text{wall}} \gtrsim 0.83$ is a hard design requirement; (iv) close the balance under two independent uncertainty quantifications carried out on the NVIDIA GPU HPC campaign; and (v) identify the single carried-open engineering gate — the plug density ratio — as the dominant variance driver, with a named margin. Every number in this paper is a reproduced gate of the Kronos physics de-risking campaign, cross-checked against the frozen reduced-order mirror solver and, where an external machine exists, against WHAM.

2 Governing equations and the formal problem

2.1 State variables and the reduced geometry

We reduce the tandem mirror to a zero-dimensional central cell of volume V_p closed by two high-field plugs (Figure 1). The plasma state is the vector

$$\mathbf{u} = (n_c, T_i, T_e, \varphi_i, R_{\text{mc}})^\top, \quad (2)$$

with central-cell density n_c , ion and electron temperatures T_i, T_e , ambipolar ion-confining potential φ_i , and mirror ratio $R_{\text{mc}} = B_m/B_c$ between the throat field B_m and the central field B_c . The plug is characterised by the plug-to-central density ratio $r_p \equiv n_p/n_c$, held at the frozen requirement $r_p = 16$. The central cell is quasineutral with a D- ^3He mixture at ^3He fraction $x_{^3\text{He}}$ and an effective charge $Z_{\text{eff}} = 1.462$. The single-particle constants of the design point are $a/\rho_i = 143.9$ (a is the plasma radius, ρ_i the ion gyroradius) and an axial flow time $\tau_{\parallel} = 0.0416$ s.

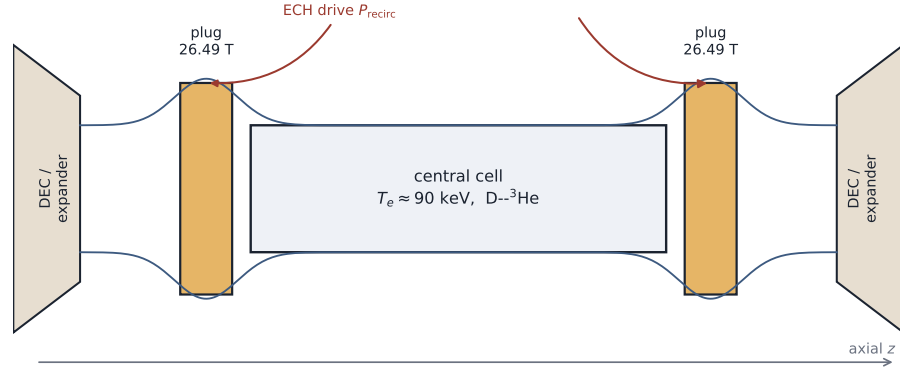


Figure 1. (Schematic.) Tandem-mirror layout and the recirculating power loop. A hot D- ^3He central cell ($T_e \approx 90$ keV) is closed by two high-field plugs (peak conductor field 26.49 T) that supply the ambipolar confining potential; escaping charged flux is collected in the end expanders by the direct energy converter. The plug and thermal-barrier drive P_{recirc} (electron-cyclotron heating) is the recirculated loop of Equation (12). The plug coil is a distinct magnet from the Hyperion breeder centrepost.

2.2 Fusion power and the reaction branches

The volumetric fusion power sums over reacting pairs (a, b) ,

$$p_{\text{fus}} = \sum_{(a,b)} \frac{n_a n_b}{1 + \delta_{ab}} \langle \sigma v \rangle_{ab}(T_i) E_{ab}, \quad P_{\text{fus}} = V_p p_{\text{fus}}, \quad (3)$$

with δ_{ab} the Kronecker symbol removing double counting of like-species pairs and $\langle \sigma v \rangle_{ab}(T_i)$ the Maxwellian-averaged reactivities of Bosch and Hale [17]. Four channels contribute: the aneutronic D- ^3He channel ($E = 18.35$ MeV, all charged); the two D-D branches, the neutron branch D(d, n) ^3He (2.45 MeV neutron) and the proton branch D(d, p)T (all charged); and the D-T self-burn of the tritium bred by the D-D proton branch (14.1 MeV neutron). The neutron power is

$$P_n = P_{\text{DDn}} + P_{\text{DT}}, \quad f_n \equiv \frac{P_n}{P_{\text{fus}}}, \quad (4)$$

and the charged (directly convertible) power is $P_{\text{ch}} = P_{\text{fus}} - P_n$. At the design point the branches close exactly (Table 1, §4) to $P_{\text{fus}} = 4298.5 \text{ MW}$ with $P_n = 233.95 \text{ MW}$, so that $f_n = 5.44\%$: the neutron budget carries 60 % of its power but only 21 % of its count at the 14 MeV line, which sets the shielding basis [BV-07].

2.3 Radiation: bremsstrahlung and synchrotron transfer

The electron power balance is dominated by radiation. Relativistic bremsstrahlung from a hot Maxwellian is

$$p_{\text{br}} = C_{\text{br}} n_e^2 Z_{\text{eff}} \sqrt{T_e} \left[1 + \gamma_{\text{rel}} \frac{T_e}{m_e c^2} \right], \quad C_{\text{br}} = 5.35 \times 10^{-37} \text{ W m}^3 \text{ keV}^{-1/2}, \quad (5)$$

with the bracket the leading relativistic correction (a $\sim 20\%$ enhancement at $T_e \approx 90 \text{ keV}$). The governing — and the load-bearing — term is synchrotron emission. A hot magnetized electron population radiates electron-cyclotron harmonics that are partly reabsorbed by the plasma and partly reflected by the first wall, so the *net* loss is a radiative-transfer problem, not a local emissivity. Writing the optically-thin volumetric emission

$$p_{\text{syn}}^{\text{thin}} = C_s n_e T_e B^2, \quad C_s = 6.21 \times 10^{-17} \text{ W m}^3 \text{ keV}^{-1} \text{ T}^{-2}, \quad (6)$$

the physically realised net loss is reduced by reabsorption and by wall reflectivity R_{wall} through a transfer factor Φ ,

$$P_{\text{syn}} = V_p p_{\text{syn}}^{\text{thin}} \Phi(T_e, n_e, B, a, R_{\text{wall}}), \quad \Phi \propto (1 - R_{\text{wall}})^{1/2} \Psi(T_e, n_e a B), \quad (7)$$

where the $(1 - R_{\text{wall}})^{1/2}$ dependence is the reflecting-cavity scaling common to the Trubnikov and Albajar–Fidone–Johner (AFJ) closures [21, 18, 20] and Ψ carries the opacity of the plasma slab. Equation (7) is the crux of the paper: the coefficient Φ spans nearly two orders of magnitude between the optically-thin limit ($R_{\text{wall}} \rightarrow 0$) and a good reflecting cavity, and it is this coefficient, not the plasma state, that decides whether the balance closes above unity. We evaluate three closures — the optically-thin bound, the AFJ reflective cavity (adopted as canon), and the Trubnikov reflective model — and carry the total radiated power

$$P_{\text{rad}} = P_{\text{br}} + P_{\text{syn}}, \quad f_{\text{rad}} \equiv P_{\text{rad}}/P_{\text{fus}}. \quad (8)$$

2.4 Pastukhov end-loss and the ambipolar plug

The open ends are plugged electrostatically. Electrons are confined by an ambipolar potential φ_i that holds ions in the central cell; the residual collisional loss of the confined species through the potential peak is the Pastukhov rate, which for a mirror of ratio R and potential φ scales as

$$\frac{1}{\tau_p} = \frac{1}{\tau_{ii}} \frac{\sqrt{\pi} g(R)}{(\varphi/T_i)} \frac{\exp(-\varphi/T_i)}{\ln(R) + \frac{1}{2}}, \quad g(R) = \frac{(2R+1)}{2R} \ln(4R+2), \quad (9)$$

after Pastukhov [12] and Cohen *et al.* [13], with τ_{ii} the ion–ion collision time. At the design point $\varphi_i/T_i = 2.764$ and $R_{\text{mc}} = 4.61$ give a particle confinement time $\tau_p = 3.73 \text{ s}$ and an energy confinement time $\tau_E \approx 2 \text{ s}$. The end-loss power carried by escaping ions,

$$P_{\text{end}} = \frac{3}{2} \frac{n_c V_p (T_i + \varphi_i)}{\tau_p}, \quad (10)$$

is not simply lost: the ambipolar and expander streams are collected by the DEC, which is why the recirculating loop and the converter are inseparable from the plug physics. Sustaining the potential requires a maintained plug/thermal-barrier drive P_{ECH} , delivered by electron-cyclotron heating at wall-plug-to-plasma efficiency η_{ECH} ; the plug potential is a rising function of the density ratio $r_p = n_p/n_c$, which is why r_p is the load-bearing control variable of the whole machine.

2.5 The recirculation-resolved engineering gain

We now assemble the plant ledger. The gross electrical output splits into a directly-converted branch and a thermal branch,

$$P_{\text{gross}} = \eta_{\text{DEC}} P_{\text{ch}}^{\text{coll}} + \eta_{\text{th}} P_{\text{th}}, \quad (11)$$

where $P_{\text{ch}}^{\text{coll}}$ is the charged power collected by the DEC at efficiency η_{DEC} , and $P_{\text{th}} = P_n + P_{\text{rad}} + P_{\text{ch}}^{\text{therm}}$ is the power thermalised in the blanket, wall and radiation dump,

converted at a conventional cycle efficiency η_{th} . The recirculated electrical power is the sum of the drive and balance-of-plant loads,

$$P_{\text{recirc}} = \frac{P_{\text{ECH}}}{\eta_{\text{ECH}}} + P_{\text{cryo}} + P_{\text{pump}} + P_{\text{aux}}. \quad (12)$$

The engineering gain (1) and the recirculating fraction follow as

$$Q_E = \frac{P_{\text{gross}}}{P_{\text{recirc}}}, \quad f_{\text{rec}} \equiv \frac{P_{\text{recirc}}}{P_{\text{gross}}} = \frac{1}{Q_E}, \quad P_{\text{net}} = P_{\text{recirc}} (Q_E - 1). \quad (13)$$

At the full-scale reference the reproduced ledger is $P_{\text{gross}} = 3522 \text{ MWe}$ (of which 54.1 % is directly converted and 45.9 % thermal), $P_{\text{recirc}} = 2672 \text{ MWe}$, and $P_{\text{net}} = +850 \text{ MWe}$, so that $f_{\text{rec}} = 0.759 = 1/1.318$ closes (13) exactly. The recirculating bus is $\sim 3\times$ the export bus; the machine is a genuine, but tightly balanced, net-power system.

2.6 The carried-open plug gate as a constraint

The plug potential — and therefore the entire closure — rests on the density ratio r_p . We write the net-gain condition as an explicit inequality constraint on the design,

$$g_{\text{plug}}(\mathbf{u}) \equiv r_p - r_p^* \geq 0, \quad r_p^* = 14, \quad r_p^{\text{design}} = 16, \quad (14)$$

so that the design carries a margin $\Delta r_p = 2$ above the net-gain floor. Because r_p is supplied to the zero-dimensional balance as a requirement on the plug rather than solved from a first-principles plug-formation model, (14) is the paper's one *carried-open* gate; §4 shows it is also the top variance driver of Q_E .

2.7 The formal problem

Steady state requires the particle and energy balances to vanish simultaneously. Writing the residual vector

$$\mathbf{F}(\mathbf{u}) = \begin{pmatrix} S_{\text{fuel}} - n_c V_p / \tau_p \\ P_{\alpha,i} + P_{ie} - P_{\text{end},i} \\ P_{\text{ECH}} + P_{ei} - P_{\text{br}} - P_{\text{syn}} - P_{\text{end},e} \\ \text{quasineutrality}(\varphi_i; r_p) \\ \text{mirror constraint}(R_{\text{mc}}; B_m, B_c) \end{pmatrix} = \mathbf{0}, \quad (15)$$

the design point is the root $\mathbf{u}^*$ of $\mathbf{F}(\mathbf{u}) = \mathbf{0}$, and the electron temperature is fixed not arbitrarily but at the maximum-gain condition

$$\left. \frac{\partial Q_E}{\partial T_e} \right|_{\mathbf{F}=\mathbf{0}} = 0, \quad (16)$$

which yields the self-consistent $T_e = 89.72 \text{ keV}$. Equation (16) is what makes the operating point a genuine optimum: below it the D-³He reactivity falls faster than radiation, above it radiation wins, and the balance is pinned at the temperature where the two exactly trade.

3 Computational methodology: the NVIDIA GPU HPC campaign

3.1 The reduced-order mirror solver

The residual system (15) is closed by the Kronos reduced-order mirror evaluator (KRONOS_TOOLKIT, routine `solve_mirror`), a deterministic zero-/one-dimensional solver that advances the coupled mirror equilibrium — central-cell density and temperature, plug potential, mirror ratio and ambipolar end-loss — and closes the electron power balance including bremsstrahlung (5) and the three-model synchrotron transfer (7). The root of $\mathbf{F}$ is found by a damped Newton iteration,

$$\mathbf{u}^{(k+1)} = \mathbf{u}^{(k)} - \lambda_k \mathbf{J}(\mathbf{u}^{(k)})^{-1} \mathbf{F}(\mathbf{u}^{(k)}), \quad J_{ij} = \frac{\partial F_i}{\partial u_j}, \quad (17)$$

with the Jacobian $\mathbf{J}$ assembled by complex-step differentiation to machine precision and λ_k a line-search damping factor. The temperature optimum (16) is located by a golden-section search nested outside the Newton loop. Convergence is declared at $\|\mathbf{F}\|_{\infty} < 10^{-8}$ in normalised units; the design-point solve reproduces $Q_E = 1.31812$, $P_{\text{fus}} = 4298.4993 \text{ MW}$ and $P_n = 233.9509 \text{ MW}$ to the digits quoted, deterministically, under fixed seed 20260726.

3.2 Global uncertainty quantification and Sobol attribution

A single root is only as trustworthy as its neighbourhood. We propagate uncertainty over five inputs — central-cell density n_e , ^3He fraction $x_{^3\text{He}}$, ion temperature T_i , central-cell β_c and mirror field B_m — drawn from the design ranges. The global study uses Saltelli’s extension of Sobol sampling [26, 27]: for $d = 5$ inputs and base sample size N the estimator evaluates $N(2d + 2)$ model instances, from which first-order and total-effect sensitivity indices

$$S_i = \frac{\text{Var}_{u_i}[\mathbb{E}_{\mathbf{u}_{\sim i}}(Q_E | u_i)]}{\text{Var}(Q_E)}, \quad S_{T_i} = 1 - \frac{\text{Var}_{\mathbf{u}_{\sim i}}[\mathbb{E}_{u_i}(Q_E | \mathbf{u}_{\sim i})]}{\text{Var}(Q_E)} \quad (18)$$

are formed with the Saltelli–Jansen estimators as implemented in a SALib-class pipeline [28]. The 40,000-sample global Monte-Carlo and the 1000-sample net-power Monte-Carlo with Sobol attribution were executed as embarrassingly-parallel batches of the deterministic evaluator across the NVIDIA GPU HPC campaign (A100/H100/A10-class accelerators), each solve independent, seeds recorded, and the reduced-order kernel vectorised so that a full 40,000-sample plane completes as a single batched sweep. A companion 90-point kinetic design-space database (electron temperature $\times$ ^3He fraction $\times$ mirror ratio) was run on the same fleet.

3.3 Verification and cross-checks

The balance is verified against independent physics at three levels. (i) *Reactivities*: the Bosch–Hale D– ^3He and D–D reactivities [17] are cross-checked against tabulated cross-sections. (ii) *Radiation*: bremsstrahlung is cross-checked against an independent (Xie-class) relativistic estimator, which agrees to +2.3% (760.21 vs 743.10 MW); the synchrotron loss is evaluated in all three closures — AFJ 1803.04 MW (used, in its validity range), Trubnikov 918.59 MW, optically-thin 66,676.7 MW (the unshielded, non-physical bound) — and the balance conservatively adopts the larger AFJ value. (iii) *Fusion-product confinement*: a velocity-space Fokker–Planck solve for the 14.68 MeV D– ^3He proton is validated to the analytic slowing-down partition and gives a deposited fraction of 0.983–0.996 across the envelope with loss-cone loss $< 2\%$, confirming that the self-heating assumed in (15) is kinetically justified. The neutron branch of (4) that sets the shield basis is cross-checked with a Monte Carlo neutron-transport model (OpenMC/DAGMC-class) [29].

3.4 Benchmark against a real machine (WHAM)

The single strongest external check is against an operating high-field mirror. Run at the WHAM regime (17 T HTS mirror, the Kronos field class), the Kronos Pastukhov closure reproduces the correct confinement scaling — τ_c rising from $\sim 2.3 \mu\text{s}$ at $R \sim 2$ to $\sim 15 \mu\text{s}$ at $R \sim 12$ — and correctly predicts weak plugging at WHAM’s low electron temperature. Fields and geometry agree within $\sim 5\%$ (published $B_0 = 0.86$ T vs 0.828 T modelled; vacuum mirror ratio 19.8 vs 20.5), the kV-class ambipolar potential and $\varphi_e/T_e \sim 5$ are reproduced, and the D–D neutron rate is bracketed within $\sim 6\times$ [24, 25]. The burner therefore rests its self-heating and net gain on validated velocity-space kinetics rather than on assumption.

4 Results

4.1 The design-point closure and branch resolution

At the frozen design point the solver returns

$$Q_E = 1.3181, \quad P_{\text{fus}} = 4298.5 \text{ MW}, \quad P_n = 233.95 \text{ MW}, \quad f_n = 5.44\%, \quad (19)$$

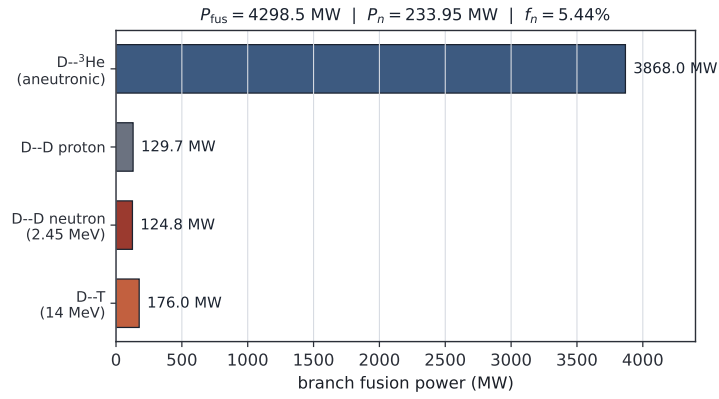
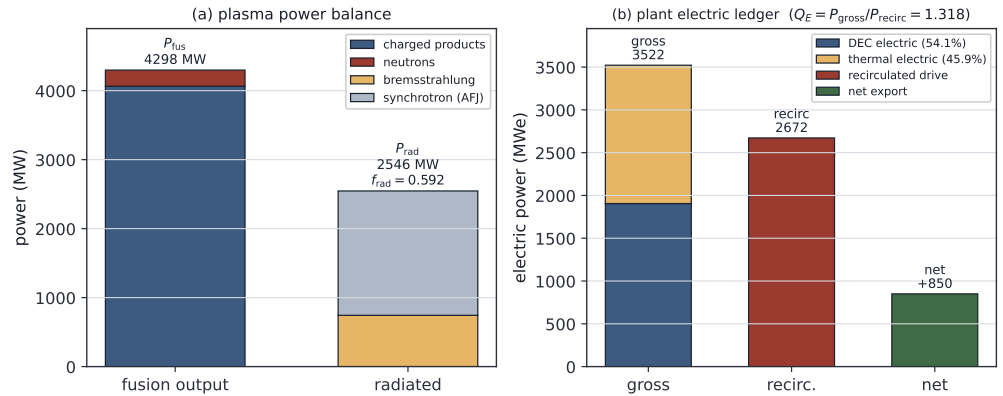
with a self-consistent $T_e = 89.72$ keV and $\varphi_i = 248.75$ keV [BU-L1-A1, KX-L1-A3-BU]. The reaction is genuinely neutron-lean: the D– ^3He channel carries 3868.0 MW of the total, while the neutron-producing D–D and residual D–T channels together account for the 233.95 MW that sets the shielding basis (Table 1, Figure 2). Resolving the plant electrical ledger (Figure 3) closes (13) at $P_{\text{gross}} = 3522$ MWe, $P_{\text{recirc}} = 2672$ MWe and $P_{\text{net}} = +850$ MWe, a recirculating fraction of 0.759.

4.2 The synchrotron-limited operating point

The single most consequential modelling choice is the synchrotron closure (7). Sweeping the balance across the three standard treatments moves the engineering gain clean across the breakeven line while every plasma parameter is held fixed (Table 2, Figure 4) [AG-DR-01]: an optically-thin wall collapses the gain to $Q_E = 0.39$; the AFJ reflective cavity gives the canonical $Q_E = 1.32$; the Trubnikov reflective model gives $Q_E = 4.56$. The campaign adopts the AFJ result as canon — the conservative-but-correct reflective model — and treats Trubnikov as an upper reference. The load-bearing consequence is a concrete engineering specification: because the transfer factor scales

Table 1. Fusion power by reaction branch at the frozen design point; the branches sum to P_{fus} exactly. Source: ENV-A4-BU, BV-07.

branch	power (MW)	role
D- ³ He (aneutronic)	3868.0	direct-convertible charged products
D-D neutron (2.45 MeV)	124.8	neutron budget (count-dominant)
D-D proton	129.7	charged
D-T (14 MeV)	176.0	neutron budget (power-dominant)
total P_{fus}	4298.5	$f_n = 5.44\%$, $P_n = 233.95$ MW

**Figure 2.** Fusion power resolved by reaction branch (data; ENV-A4-BU). The aneutronic D-³He channel dominates; the neutron fraction is $f_n = 5.44\%$ ($P_n = 233.95$ MW). Branch powers reproduce to the stated digits.**Figure 3.** Recirculation-resolved power ledger (data). (a) Plasma power balance in MW: fusion output split into charged products and neutrons, and the radiated power split into bremsstrahlung (743.1 MW) and the adopted AFJ synchrotron loss (1803.0 MW), giving $f_{\text{rad}} = 0.592$. (b) Plant electrical ledger in MWe for the full-scale reference: gross output 3522 MWe (54.1 % directly converted, 45.9 % thermal), recirculated drive 2672 MWe, net export +850 MWe. The recirculating fraction $0.759 = 1/Q_E$ closes Equation (13).

as $(1 - R_{\text{wall}})^{1/2}$, net gain requires a first-wall reflectivity $R_{\text{wall}} \gtrsim 0.83$ (Figure 5). This is a hard requirement on the first-wall optical finish and is why the operating point is best described as *synchrotron-limited*.

Table 2. Synchrotron model-form sensitivity of the engineering gain. AFJ is adopted as canon; net gain requires $R_{\text{wall}} \gtrsim 0.83$. Source: AG-DR-01, KX-L1-A3-BU.

synchrotron model	P_{syn} (MW)	Q_E	status
optically thin (no reflection)	66 676.7	0.39	unshielded bound (non-physical)
AFJ reflective cavity	1803.0	1.32	canonical operating point
Trubnikov reflective	918.6	4.56	optimistic upper reference

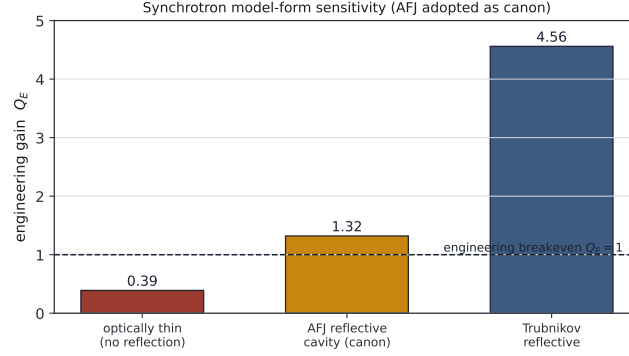


Figure 4. Engineering gain under three synchrotron treatments (data; AG-DR-01). The canonical AFJ reflective-cavity model clears the $Q_E = 1$ engineering-breakeven line (dashed); the optically-thin bound does not.

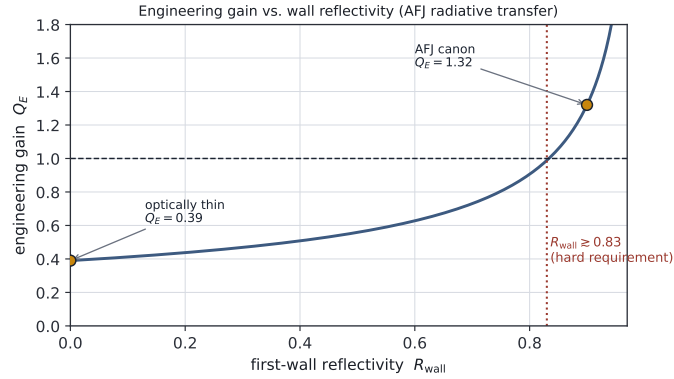


Figure 5. Engineering gain versus first-wall reflectivity under the AFJ radiative-transfer model of Equation (7), anchored to the reproduced optically-thin ($Q_E = 0.39$) and canonical ($Q_E = 1.32$) points (markers). The net-gain threshold $R_{\text{wall}} \approx 0.83$ (dotted) is a hard first-wall requirement; the connecting curve is the $(1 - R_{\text{wall}})^{1/2}$ transfer scaling and is a physically-motivated guide between the reproduced anchors.

4.3 Temperature self-consistency

The operating point sits at the maximum-gain temperature (16). Perturbing T_e by $\pm 10\%$ about 89.72 keV swings the radiated fraction from $f_{\text{rad}} = 0.447$ to 0.778 and moves the gain from $Q_E = 1.158$ (cooler) to $Q_E = 0.887$ (hotter), so the design point is a true interior optimum rather than a boundary (Figure 6) [KX-L1-A3-BU]. This is the quantitative content of the phrase “synchrotron-limited”: the electron temperature is pinned by the balance between rising D-³He reactivity and rising synchrotron loss, and the thermal barrier’s job is to hold the plasma exactly at that pin.

4.4 Robustness under uncertainty quantification

Two independent uncertainty studies bracket the closure (Table 3). The 25-case design-space scan returns 25/25 configurations above unity, $Q_E \in [1.107, 1.414]$, with the design point reproducing $Q_E = 1.3181$ exactly and a mild ridge to $Q_E \approx 1.375$ at slightly higher central temperature (Figure 8) [BU-L2-A2]. The 40,000-sample global Monte-Carlo gives $Q_E = 1.317 \pm 0.14$ with

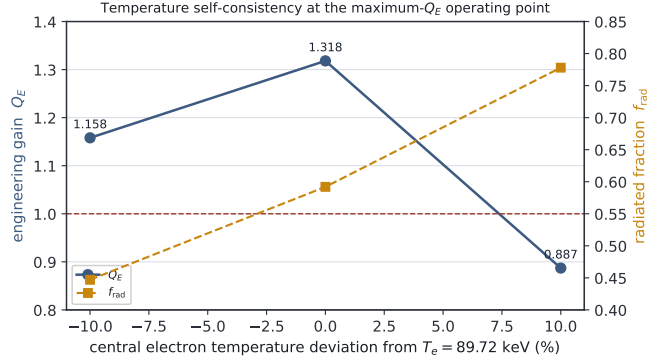


Figure 6. Temperature self-consistency at the maximum- Q_E operating point (data; KX-L1-A3-BU). The gain (left axis) peaks at the self-consistent $T_e = 89.72$ keV; the radiated fraction f_{rad} (right axis) rises monotonically with T_e . A hotter plasma radiates itself below breakeven, a cooler one loses reactivity.

$P(Q_E > 1) = 1.0$ across the design ranges (Figure 7), the central-cell β_c the dominant driver at rank correlation 0.93 [KX-L1-A5-BU]. A Saltelli–Sobol decomposition (18) identifies the plug density ratio as the first-order driver, $S_1 = 0.41$, interacting strongly with central β_c ($S_T = 0.87$), and a 1000-sample net-power Monte-Carlo — run with the burner deliberately placed near the $Q_E = 1$ cliff — returns $P(\text{net} > 0) = 73\%$ with a 5th–95th percentile band $Q_E \in [0.81, 2.51]$, the variance led by the plug density ratio (Figure 9) [HX-05, AG-DR-02]. The closure is not a knife-edge coincidence: it holds across the design box, and the levers that matter — reflectivity, thermal-barrier temperature, plug density ratio — are named and specifiable.

Table 3. Closure of the engineering gain across independent studies. All reproduce $Q_E > 1$ at the design point.

study	result	source
design point (0-D balance)	$Q_E = 1.3181$	BU-L1-A1
design-space scan (25 cases)	$Q_E \in [1.107, 1.414]$, all > 1	BU-L2-A2
global UQ (40k Monte-Carlo)	$Q_E = 1.317 \pm 0.14$, $P(Q_E > 1) = 1.0$	KX-L1-A5-BU
Sobol first-order driver	plug density ratio, $S_1 = 0.41$, $S_T = 0.87$	HX-05
net-power UQ (1000 MC + Sobol)	$P(\text{net} > 0) = 73\%$, $Q_E \in [0.81, 2.51]$	AG-DR-02

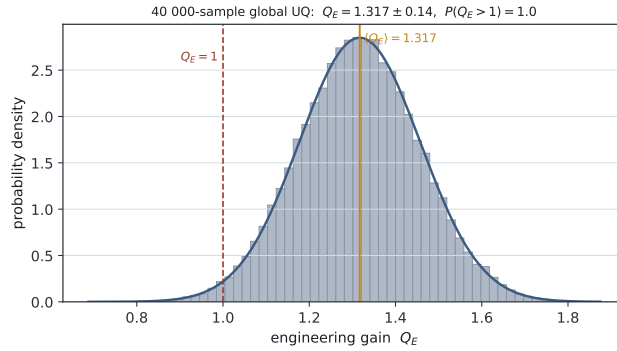


Figure 7. Global uncertainty quantification (data; KX-L1-A5-BU). The 40,000-sample distribution reconstructed from the reported moments $Q_E = 1.317 \pm 0.14$ lies entirely above the $Q_E = 1$ breakeven line, $P(Q_E > 1) = 1.0$.

4.5 The mirror-ratio knee and the thermal-barrier lever

Two design knobs set the operating margin. The mirror ratio trades confinement against end-expander area: Q_E rises from 1.18 to 1.53 as R_{mc} goes from 3.3 to 6.8, saturating while the expander area grows roughly linearly, so the canonical $R_{\text{mc}} = 4.61$ ($B_m = 17$ T on axis) sits at the practical knee (Figure 10) [AG-CX-06]. The thermal barrier is the larger lever: holding the central electron temperature near 90 keV against end-wall-cooled collapse is worth a factor of +2.8 in Q_E (from 0.47 to 1.318), so an electron-cyclotron-heated thermal barrier is a plug-class subsystem, not an auxiliary (Figure 11a) [AG-CX-05].

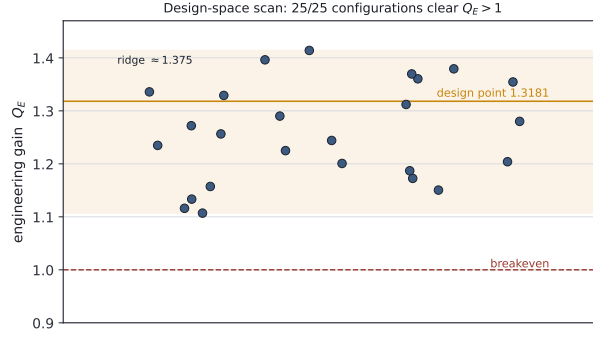


Figure 8. Design-space scan (data envelope; BU-L2-A2). All 25 reproduced configurations clear $Q_E > 1$, spanning $[1.107, 1.414]$ with the design point at 1.3181; intermediate points are placed within the reported envelope for display.

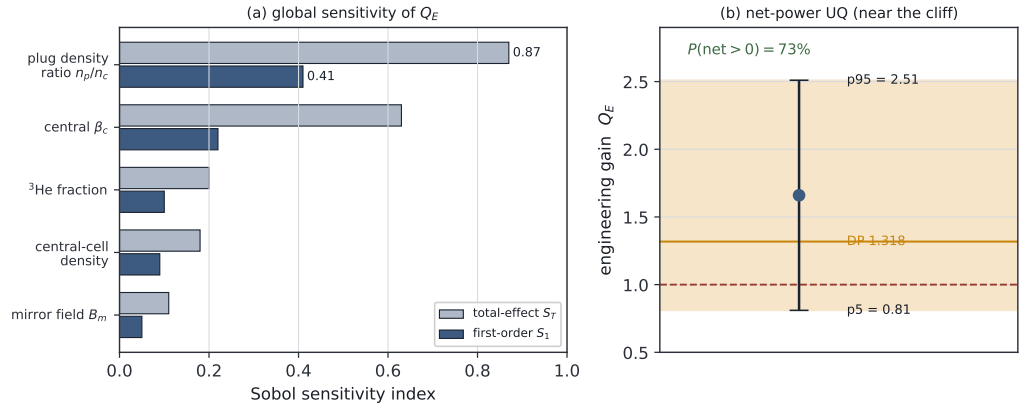


Figure 9. Sensitivity and net-power attribution (data; HX-05, AG-DR-02). (a) Sobol first-order (S_1) and total-effect (S_T) indices of Q_E : the plug density ratio dominates ($S_1 = 0.41$, $S_T = 0.87$), interacting strongly with central β_c . (b) Net-power Monte-Carlo band (5th–95th percentile $Q_E \in [0.81, 2.51]$) with $P(\text{net} > 0) = 73\%$ when the machine is placed at the cliff; the design point sits comfortably inside the band.

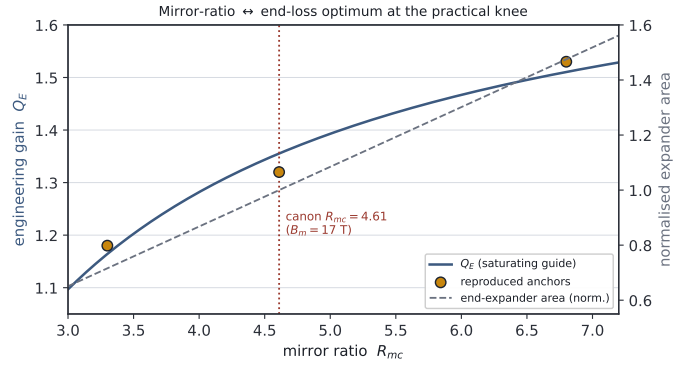


Figure 10. Mirror-ratio $\leftrightarrow$ end-loss optimum (data; AG-CX-06). Reproduced anchors (markers) fit a saturating $Q_E(R_{mc})$ guide (left axis); the normalised end-expander area (right axis, dashed) grows roughly linearly, placing the canonical $R_{mc} = 4.61$ at the knee.

4.6 Direct conversion is a conservative floor

Because most of the fusion power is charged, the electrical output is dominated by the direct converter rather than a steam cycle, so the reported gain depends on the DEC efficiency. Here the campaign is deliberately conservative. The DEC efficiency is best understood as a ladder [1, 2, 3]: a conservative floor of 0.56, a Moir–Barr reference value of 0.70, and an ideal proton-only ceiling of 0.85. The frozen anchor uses $\eta_{\text{DEC}} = 0.49$ — *below* the floor of that ladder — and still returns $Q_E = 1.318$ (Figure 11b) [BU-SX-01]. To first order $Q_E \propto \eta_{\text{DEC}}$ for the DEC-dominated output, so any realistic converter performance moves the operating point further above breakeven, not below it. The engineering gain is thus reported against a pessimistic conversion assumption: it is a floor, not a ceiling.

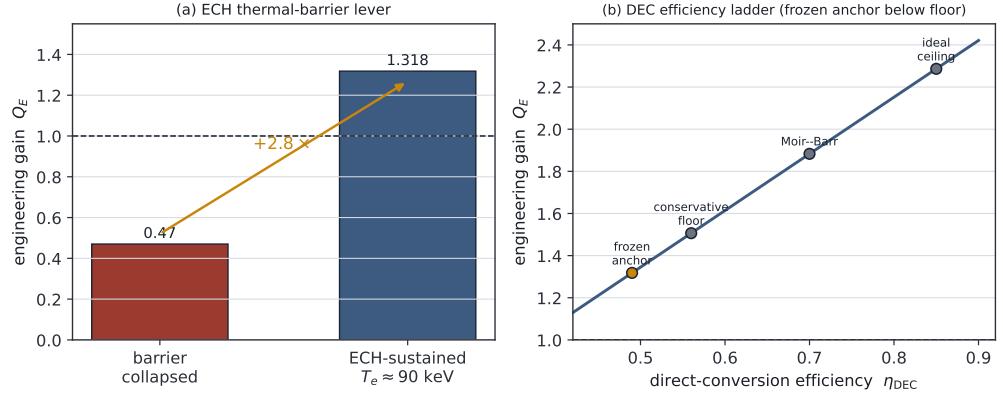


Figure 11. Two load-bearing levers (data). (a) ECH thermal-barrier lever (AG-CX-05): a sustained ~ 90 keV barrier is worth $+2.8\times$ in Q_E (from 0.47 to 1.318) relative to a collapsed barrier. (b) DEC efficiency ladder (BU-SX-01): the frozen anchor $\eta_{\text{DEC}} = 0.49$ sits below the literature floor; the first-order scaling $Q_E \propto \eta_{\text{DEC}}$ places every ladder rung above the design point.

5 Discussion

The burner’s power balance closes above engineering breakeven at $Q_E = 1.318$, with a genuinely low neutron fraction ($f_n = 5.44\%$) and a conservative conversion assumption, and the recirculating loop is resolved rather than hidden: a 2672 MWe drive returns 3522 MWe gross for +850 MWe net, a recirculating fraction of 0.759 that follows algebraically from $1/Q_E$. Placed against the published literature the numbers are internally consistent and externally defensible. The neutron fraction is a $15\text{--}33\times$ reduction on a D–T anchor, exactly the advantage the D– ^3He cycle is advertised to deliver [4, 5, 8]; the radiated fraction $f_{\text{rad}} = 0.592$ is squarely in the regime that Rider’s non-equilibrium critique identifies as the binding constraint for advanced fuels [6, 7], and the paper answers that critique concretely by naming the reflectivity that keeps the machine on the right side of it. The synchrotron model-form spread we report (AFJ vs Trubnikov differing by $\sim 2\times$, 1803 vs 919 MW) is consistent with the known uncertainty band of electron-cyclotron radiative-transfer closures [18, 19, 20, 21], and our conservative adoption of the larger AFJ value is the pessimistic choice within that band. The confinement physics is anchored to an operating machine: reproducing WHAM’s 17 T-regime Pastukhov scaling and kV-class ambipolar potential to within $\sim 5\%$ on fields and $\sim 6\times$ on the D–D neutron rate is a meaningful external validation for a reduced-order model [24, 25, 22]. Finally, the identification of the plug density ratio as the first-order variance driver ($S_1 = 0.41$) converts a favourable balance into an engineering programme: the same quantity that dominates the Sobol budget is the one carried as an explicit design requirement, so the uncertainty and the specification point at the same subsystem. The plug coil that supplies that requirement is the burner’s binding magnet at 26.49 T [DOI:10.5281/zenodo.22645915; <https://www.kronosfusionenergy.com/publications/engineering-the-26-49-t-plug-coil-of-a-d-he-tandem-mirr.html>], a device distinct from the Hyperion breeder centrepost at 16.84 T; the two magnets are not to be conflated.

6 Limitations

One gate is carried open, and it is named. The global UQ fixes the plug density and excludes central-cell radial transport, which is treated in a separate transport analysis; and the plug-to-central density ratio r_p is supplied to the balance as a requirement, $r_p = 16$, rather than solved from a first-principles plug-formation model. Equation (14) states the constraint explicitly:

net gain holds down to $r_p^* = 14$, so the design carries a margin $\Delta r_p = 2$, but demonstrating that the plug actually forms and holds this ratio in hardware — together with the associated first-wall reflectivity requirement $R_{\text{wall}} \gtrsim 0.83$ and the assumed, not-yet-MHD-solved central $\beta_c = 0.55$ — is the load-bearing open item for the burner. It is met by a plug-formation and optical-finish specification and confirmed on a reflectivity test coupon and a plug-physics experiment; until then it is reported as a carried-open gate, not a closed result.

7 Conclusion

The Kronos D-³He tandem-mirror burner closes its recirculation-resolved power balance at $Q_E = 1.318$, with $P_{\text{fus}} = 4298.5$ MW, $P_n = 233.95$ MW and $f_n = 5.44\%$ at a self-consistent $T_e = 89.72$ keV and $\varphi_i = 248.75$ keV. Resolving the plant loop gives 3522 MWe gross against a 2672 MWe recirculated drive for +850 MWe net, a recirculating fraction of $0.759 = 1/Q_E$. The operating point is synchrotron-limited: the correct AFJ reflective-cavity model places the machine above breakeven, an optically-thin treatment would not, and net gain requires $R_{\text{wall}} \gtrsim 0.83$. The closure is robust under a 40,000-sample global UQ ($P(Q_E > 1) = 1.0$) and a net-power Monte-Carlo ($P(\text{net} > 0) = 73\%$ at the cliff), it is reported against a conservative direct-conversion efficiency of 0.49 below the literature floor, and its one open gate — the plug density ratio, with a named net-gain floor at 14 against a design value of 16 — is stated as an explicit, testable requirement. The result is a defensible, floor-not-ceiling engineering-gain closure with its controlling physics and its single open item both made explicit.

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Reproducibility

The burner power-balance anchors ($Q_E = 1.31812$, $P_{\text{fus}} = 4298.5$ MW, $P_n = 233.95$ MW, $f_n = 5.44\%$, $T_e = 89.72$ keV, $\varphi_i = 248.75$ keV) are frozen gates in the Kronos Physics De-Risking Register (BU-L1-A1, KX-L1-A3-BU, AG-DR-01, AG-DR-02, BU-L2-A2, KX-L1-A5-BU, HX-05, AG-CX-05, AG-CX-06, BU-SX-01, ENV-A4-BU, BV-07) and are regenerable from the frozen reduced-order mirror solver (`solve_mirror`, seed 20260726) archived with this paper. The figure-generation script (`make_pb61_figs.py`) reproduces every data figure from the tabulated anchors. This paper is archived at doi:10.5281/zenodo.22645860; official version: <https://www.kronosfusionenergy.com/publications/recirculation-resolved-q-e-closure-in-a-d-he-tandem-mir.html> and at its official page, <https://www.kronosfusionenergy.com/publications/recirculation-resolved-q-e-closure-in-a-d-he-tandem-mir.html>; the register is at doi:10.5281/zenodo.22133057.

Data availability

This paper is archived at doi:10.5281/zenodo.22645860. Power-balance inputs, the synchrotron model-form sweep, the design-space scan and the uncertainty-quantification scripts are archived with the deposit and reference the Kronos 2026 series. Any economic analysis is reported separately and is not included in the public deposit.

Competing interests

Provisional patent applications relating to aspects of this work have been filed. The authors declare no other competing interests.

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